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KADI SARVA VISHWAVIDYALAYA
B.E. Semester-VII Examination (October/ 2024)

SUBJECT CODE: CT704A-N

DATE: 23/10/2024

TIME: 12:30 pm to 3:30 pm

SUBJECT NAME: Distributed Systems

TOTAL MARKS: 70

DISTRIBUTED SYSTEMS

Instructions:

1. Attempt all Questions.
 2. Figures to the right indicate full marks.
 3. Indicate clearly, the options you attempt along with its respective Que.No.
 4. Answer each section in separate Answer sheet.
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SECTION - 1

- Q-1.** a) Write a short note on www and client-server model. 5
- b) Why are scalability and heterogeneity important features in the design of a distributed system? 5
- c) Classify and explain types of computer networks. 5
- OR**
- Q-1** c) Differentiate FLIP and VMTP protocols. 5
- Q-2.** a) Explain failure handling mechanism used in IPC. 5
- b) Which are the parameter passing semantics of RPC? Explain any one of them. 5
- OR**
- Q-2** a) Draw and explain IPC message format. 5
- b) What is RPC? Explain basic operation of RPC with diagram. 5
- Q-3.** a) Explain different message buffering strategies. 5
- b) What are Logical Clocks? Write applications of Logical Clocks. 5
- OR**
- Q-3.** a) List out various election algorithms. Explain token ring algorithm in detail. 5
- b) List and explain conditions which are necessary and sufficient for a deadlock to occur in a system. 5

P.T.O.

SECTION –2

- Q-4. a) Write importance of File Models in Distributed Systems and list types of File Models in Distributed Systems. 5
- b) What is process migration? Explain advantages of process migration. 5
- c) Explain purpose of load balancing in Distributed Systems. 5

OR

- Q-4. c) Explain different fault tolerance strategies in distributed system management. 5
- Q-5. a) What is DSM? Explain DSM architecture using nodes. 5
- b) How to achieve security in distributed systems? Explain any one in detail. 5

OR

- Q-5. a) Which are the methods of DFS lookup? Explain any one of them. 5
- b) Write a note on grid computing. 5
- Q-6. a) What is consistency model? Discuss the various consistency models used in DSM system. 5
- b) Write and explain DFS implementation. 5

OR

- Q-6. a) Write a note on service oriented architecture. 5
- b) Describe goals of Distributed System security. 5

*****ALL THE BEST*****